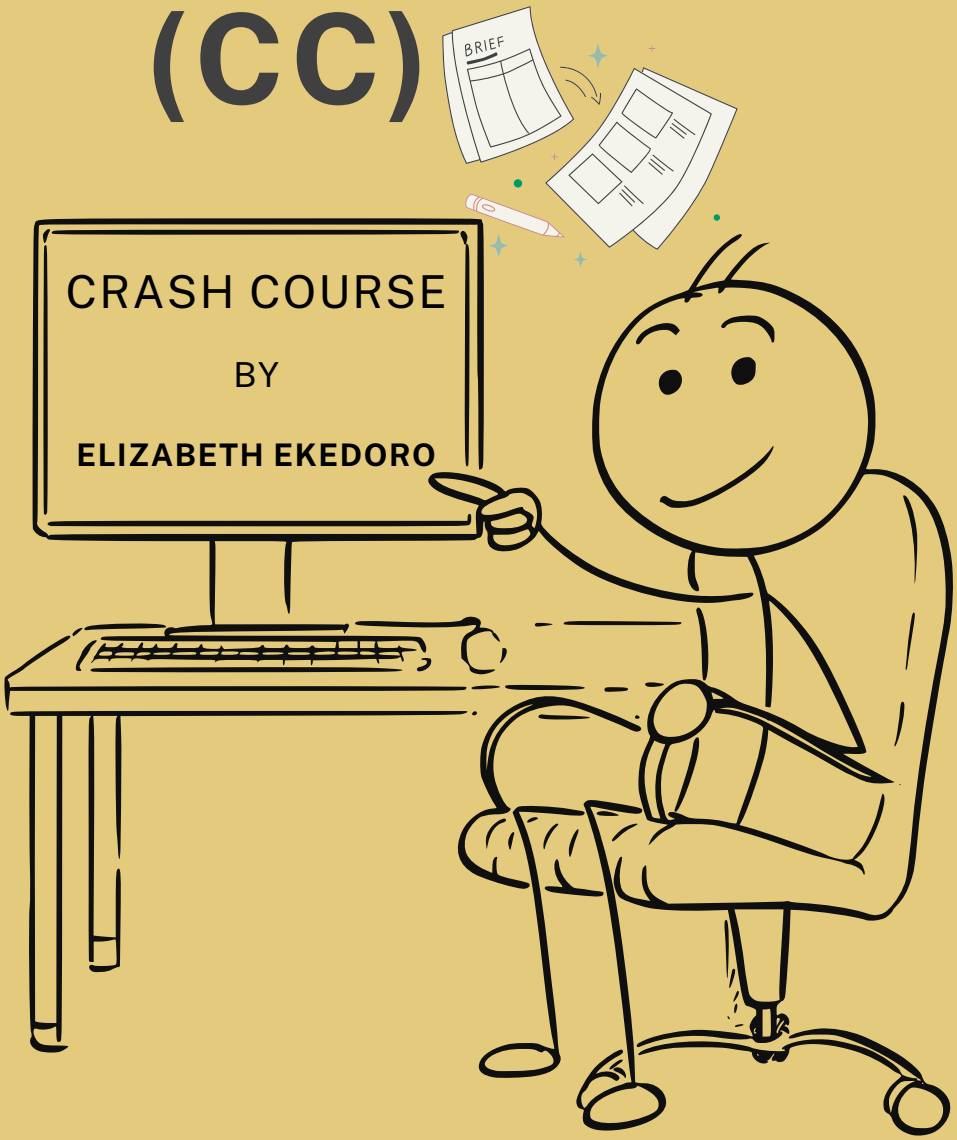
ISC2 CERTIFIED IN CYBERSECURITY (CC)



ABOUT THE INSTRUCTOR

Elizabeth Ekedoro

Elizabeth Ekedoro is a SOC Analyst and the founder of CC Champions Community (C4).

C4 is in partnership with ISC2 to provide mentorship, free training, and support for cybersecurity enthusiasts.

With hands-on experience defending real-world systems and a passion for breaking down complex topics, she's here to help you confidently step into the field of cybersecurity.





What you'll learn

This beginner-focused crash course prepares you for the ISC2 CC exam, covering all five domains and "Quick Check" questions to test your understanding as you go.

HOW TO SIGN UP

Follow these simple steps to get your free Certifed in Cybersecurity certifcation exam voucher:

- Complete your ISC2 Candidate application form using the link below and select Certified in Cybersecurity as your certification of interest.
- If you don't already have an ISC2 account, you'll need to create one using the same link below:

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DOMAIN BREAK DOWN AND WEIGHT

26%

Domain 1: Security Principles

Learn security concepts, risk management, ISC2 Code of Ethics, and governance.

10%

Domain 2: Business Continuity (BC), Disaster Recovery (DR) & Incident Response Concepts

Understand how to prepare for, respond to, and recover from incidents.

22%

Domain 3: Access Controls Concepts

Find out what you need to know about physical access and logical access controls.

24%

Domain 4: Network Security

Dive into networking, cyber threats, and defense tools.

18%

Domain 5: Security Operations

Learn all about data security, system hardening, best practice security policies and awareness training.

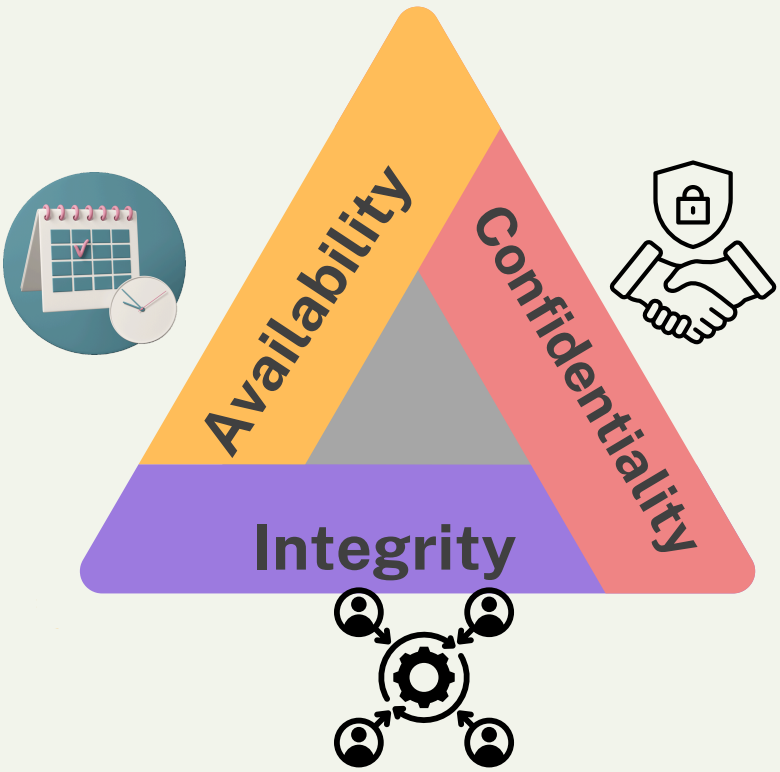
DOMAIN 1: SECURITY PRINCIPLES

Core concepts

This domain introduces foundational cybersecurity concepts for protecting information and systems.

CIA Triad: Ensures:

- Confidentiality: Only authorized userd have access (e.g., encryption).
- Integrity: Data accuracy and zero modification (e.g., hashing).
- Availability: Authorized users can access data when needed (e.g., backups).



IAAA Framework

- Identification: Claiming an identity (e.g., username)
- Authentication: Verifying identity (e.g., password).
- Authorization: Defining permissions of an authenticated user
- Accounting: Tracking user activities to ensure compliance

DOMAIN 1: SECURITY PRINCIPLES

Risk Management

Risk management involves identifying, assessing, and mitigating risks to minimize their impact.

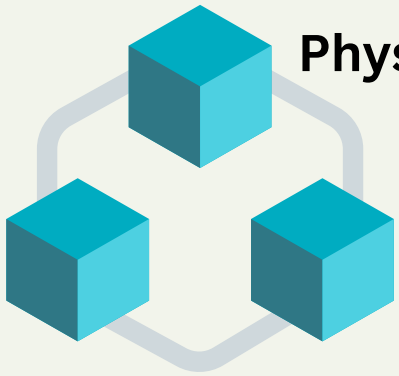
· Asset: Something in need of protection.

· Vulnerability: A gap or weakness in those protection efforts.

· Threat: Something or someone that aims to exploit a vulnerability to thwart protection efforts.

Security Controls

Security controls are physical, technical, and administrative measures designed to protect the confidentiality, integrity, and availability of information systems. They help reduce risk to an acceptable level.



Physical Controls

Technical Controls

Administrative Controls

Ethics and Governance

· The ISC2 Code of Ethics ensures professional conduct

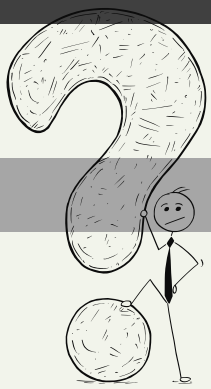
· Governance aligns cybersecurity with business goals through policies and compliance with laws like:

- PCI-DSS (Payment Card Industry Data Security Standard)
- HIPAA (Health Insurance Portability and Accountability Act) - USA
- GDPR (General Data Protection Regulation) Protects the personal data and privacy of individuals in EU.

DOMAIN 1: SECURITY PRINCIPLES



Quick Check



Question 1: An employee accidentally changes financial records. Later, the data doesn't match the backup. What principle is violated?

A. Confidentiality
B. Integrity
C. Availability
D. Authentication

Answer: B. Integrity, the data was changed, making it unreliable

Question 2: After a phishin g attack, logs show who downloaded sensitive files. Which IAAA franework helped identify this?

Which principle helped identify this?

A. Availability
B. Authorization
C. Accountability
D. Non-repudiation

Answer: C. Accountability, logs linked the actions to a user

DOMAIN 2: BUSINESS CONTINUITY, DISASTER RECOVERY INCIDENT RESPONSE CONCEPTS

Business Continuity (BC)

- Ensures critical operations continue during and after disruptions.
- Focuses on maintaining availability of services.
- Includes planning for natural disasters, cyberattacks, or system failures.

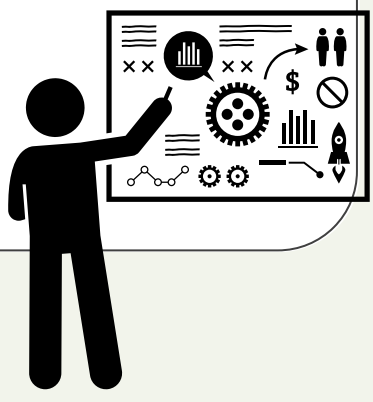


Disaster Recovery (DR)

- Focuses on restoring IT systems and data after a disruption via,
  - Mitigation: Reducing risks.
  - Preparation: Planning recovery.
  - Response: Containing damage.
  - Recovery: Restoring systems.

Incident Response (IR)

- A structured approach to detecting, responding to, and recovering from security incidents. Involves key steps:
  - Preparation: Develop policies and train staff.
  - Detection: Identify incidents via logs.
  - Response and Mitigation: Contain damage.
  - Reporting: Communicate to stakeholders.
  - Recovery: Restore systems.
  - Lessons Learned: Improve responses.



DOMAIN 2: BUSINESS CONTINUITY, DISASTER RECOVERY INCIDENT RESPONSE CONCEPTS

Business Continuity Plan (BCP)

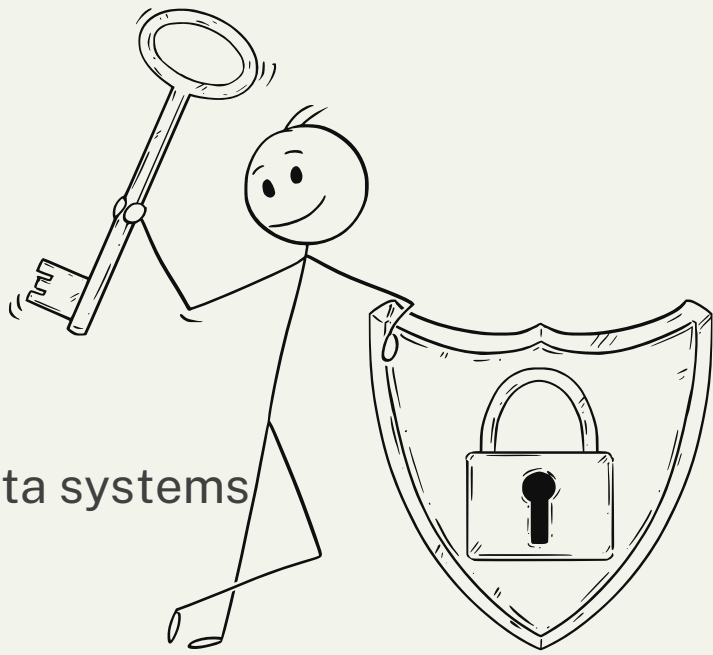
- Goal: To ensure essential business functions continue during and after a disruption. Focuses on:

- Maintaining availability of services
- Minimizing downtime
- Keeping the business operational (e.g., customer support, finance)
- Includes risk assessments, business impact analysis (BIA), and continuity strategies.

Disaster Recovery Plan (DRP)

- Goal: Restore normal operations quickly and minimize data loss.
  - Data backups
  - Alternative data centers
  - System recovery procedures

Key Difference

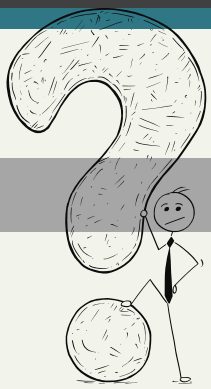


- BCP = Keeps the business running
- DRP = Recovers technology and data systems

DOMAIN 2: BUSINESS CONTINUITY, DISASTER RECOVERY INCIDENT RESPONSE CONCEPTS



Quick Check



Question 1: A ransomware attack locks a companys servers. Which department is least likely to execute the DRP?

A. IT
B. Executive
C. Marketing
D. Public Relations

Answer: C. Marketing, as DRP execution involves IT, executives, and PR

Question 2: During a cyberattack, log analysis reveals unauthorized access. Which incident management phase is this?

A. Preparation
B. Detection
C. Response
D. Recovery

Answer: B. Detection, as log analysis identifies incidents.

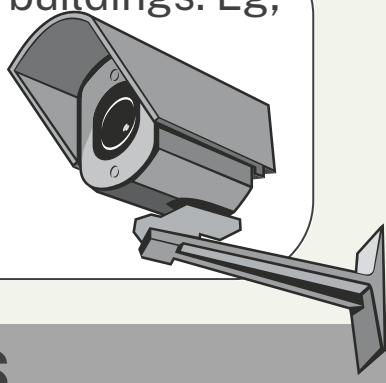
DOMAIN 3: ACCESS CONTROLS CONCEPTS

Security Controls

- This is about restricting access to systems, data, and physical areas based on who the user is, what they're allowed to do, and verifying their identity.

Physical Controls

- Controls that prevent physical access to systems or buildings. Eg,
  - Security guards/Dogs
  - Door locks/Mantraps/Turnstiles
  - Fences
  - Security cameras (CCTV)
  - Biometric scanners at entrances/Swipe Cards

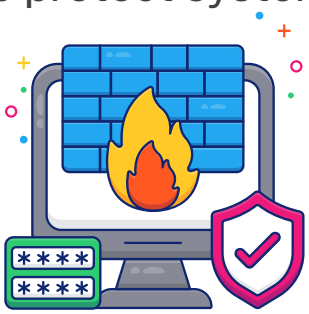


Administrative Controls

- Policies, procedures, and guidelines that define security rules and behavior. E.g,
  - Security awareness training
  - Access control policies
  - Background checks
  - Incident response plans
  - Acceptable use policies

Technical/Logical Controls

- Controls enforced by hardware or software to protect systems and data. E.g,
  - Passwords
  - Firewalls
  - Encryption
  - Access Control Lists (ACLs)
  - Multi-Factor Authentication (MFA)



DOMAIN 3: ACCESS CONTROLS CONCEPTS

Access Control Models

- Define how access to resources is granted or restricted in an information system. They determine who can access what, under which conditions. E.g,

Discretionary Access Control (DAC)

- Owner-controlled: The data owner decides who gets access.
- Flexible but less secure – users can grant others access.
- Example: A user sets file permissions on their personal folder.

Mandatory Access Control (MAC)

- System-enforced: Access is based on labels (e.g., Top Secret, Confidential).
- Used in high-security environments (e.g., military).
- Users cannot change access permissions.
- Example: Only users with “Top Secret” clearance can access “Top Secret” files.

Role-Based Access Control (RBAC)

- Access is based on a user’s role or job function.
- Roles have predefined permissions.
- Simplifies management in large organizations.
- Example: HR role can access employee records, but not server configurations.

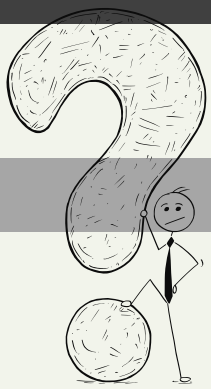
Attribute-Based Access Control (ABAC)

- Access is based on user, resource, and environment attributes.
- More precise and adapts based on real-time context.
- Example: A user can only access data if they are a manager AND it's during work hours AND they're using a company device.

DOMAIN 3: ACCESS CONTROLS CONCEPTS



Quick Check



Question 1: What is the key difference between DAC and MAC?

A. DAC is role-based; MAC is attribute-based
B. DAC is owner-controlled; MAC is authority-controlled
C. DAC prioritizes confidentiality; MAC prioritizes availability
D. DAC uses clearance levels; MAC uses policies

Answer: B. DAC allows owners to set permissions; MAC uses a central authority.

Question 2: A company uses cameras and locks to secure its data center. What type of control is this?

A. Administrative
B. Technical
C. Logical
D. Physical

Answer: D. Physical, as cameras and locks are physical barriers

DOMAIN 4: NETWORK SECURITY

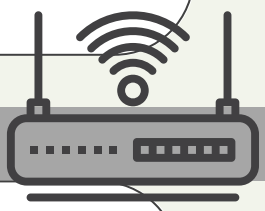
Networking

- Networking is the practice of connecting devices like computers, servers, and printers so they can share data and resources.



Types of Networks

- LAN (Local Area Network): Small, local (e.g., office or home).
- WAN (Wide Area Network): Covers large areas (e.g., the Internet).
- WLAN (Wireless LAN): Uses Wi-Fi.
- VPN (Virtual Private Network): Encrypts traffic over public networks for secure remote access.



Common Network Devices

- Router: Connects different networks; routes traffic.
- Switch: Connects devices in a LAN and manages data flow.
- Firewall: Monitors and controls incoming/outgoing traffic based on security rules.
- Endpoints: Devices like desktops, phones, tablets

Ports and Protocols

- Define how services communicate securely and efficiently. E.g,
  - 20/21 FTP File Transfer Protocol
  - 22 SSH Secure remote login
  - 25 SMTP Sending email
  - 53 DNS Domain Name System
  - 443 HTTPS Secure web traffic



DOMAIN 4: NETWORK SECURITY

OSI MODEL (7 LAYERS)

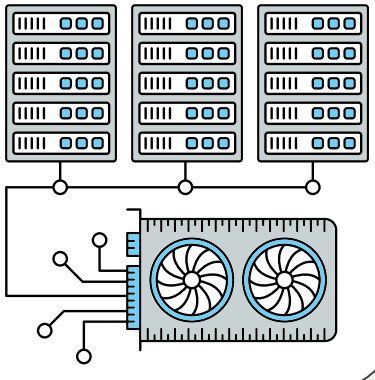
- A framework for understanding how network communication works in layers.
- Mnemonic: "All People Seem To Need Data Processing"

Layer	Name	Function
7	Application	User interaction (e.g., web browser)
6	Presentation	Data format (encryption, compression)
5	Session	Session control (start, stop, manage)
4	Transport	Reliable data transfer (TCP/UDP)
3	Network	Routing and addressing (IP, ICMP)
2	Data Link	Ensures data is delivered rightly
1	Physical	Physical media (cables, Wi-Fi,)

TCP/IP Model

- TCP/IP (Transmission Control Protocol/Internet Protocol) is the internet's delivery system that ensures messages go to the right place and arrive safely.

TCP/IP Layer	OSI Layer
Application	Layers 7–5
Transport	Layer 4
Internet	Layer 3
Network Access	Layers 2–1



DOMAIN 4: NETWORK SECURITY

Common Security Attacks

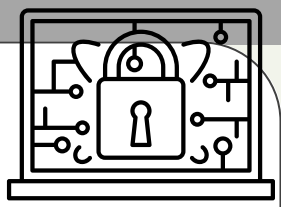
- DoS/DDoS – Overload network with traffic
- Spoofing – Pretending to be another device/user
- MITM – Intercepting data between sender & receiver



Detection & Prevention Tools:

- (Intrusion Detection System (IDS): Monitors network or system activity to detect suspicious behavior or known threats.
- Network-based IDS(NIDS): Monitors network traffic for signs of attacks (e.g., sniffing at a switch or firewall).
- Host-based IDS (HIDS): Monitors a specific device (host), checking logs, system files, and behavior for signs of compromise.
- (Security Information and Event Management (SIEM): collects, analyzes, and correlates logs and alerts from various sources like firewalls, IDS, antivirus, and endpoints.

Security Models

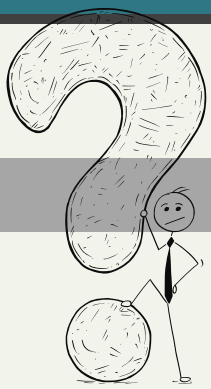


- Defense in Depth: Layered security approach.
- Zero Trust: No implicit trust; always verify everything.
- NAC (Network Access Control): Controls device access to network.

DOMAIN 4: NETWORK SECURITY



Quick Check



Question 1: An employee sends an email via a company server. Which port is used for SMTP?

A. 22
B. 25
C. 80
D. 443

Answer: B. Port 25, used for SMTP to send emails

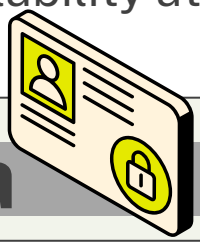
Question 2: Which OSI model layer handles physical connections like cables?

A. Application
B. Transport
C. Physical
D. Network.

Answer: C. Physical, as it deals with hardware connections

DOMAIN 5: SECURITY OPERATIONS

- This deals with how data is collected, processed, stored, shared, and disposed of securely. The goal is to protect data confidentiality, integrity, and availability at every stage of its life



Life-Cycle of Data

- Create: Data is generated or collected. Apply classification (e.g., public, confidential).
- Store: Data is saved in systems or the cloud. Use encryption and access controls.
- Use: Data is accessed or modified. Limit access to authorized users.
- Share: Data is sent to others. Use secure channels (VPN, HTTPS).
- Archive: Inactive data is stored long-term. Protect with encryption and retention policies.
- Destroy: Data is no longer needed. Securely wipe or physically destroy it.

Configuration Management

- It ensures that systems are configured securely and stay that way over time.



- Focus: Maintain secure, known setups.
- Goal: Prevent drift or unauthorized settings
- Example: Track firewall and server configurations

DOMAIN 5: SECURITY OPERATIONS

Patch Management

- The process of acquiring, testing, and deploying software updates (patches) to fix vulnerabilities or bugs in systems and applications.
  - Focus: Fix vulnerabilities or bugs
  - Goal: Strengthen security and stability
  - Example: Apply OS and software security patches



Change Management

- A structured process to request, assess, approve, implement, and document any changes to IT systems or configurations
  - Focus: Control system changes
  - Goal: Reduce risk of errors or outages
  - Example: Approve and log changes before implementation



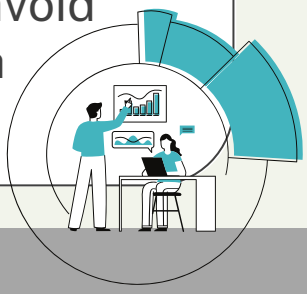
Cryptography, Encryption, Hashing

- Cryptography: The science of securing information by transforming it into an unreadable format to protect confidentiality, integrity, and authenticity.
- Encryption: Converts readable data (plaintext) into unreadable data (ciphertext) protecting data from unauthorized access.
- Symmetric Encryption: Same key for encrypt/decrypt (e.g., AES).
- Asymmetric Encryption: Different keys for encrypt/decrypt (e.g., RSA).
- Hashing: Creates a fixed-length, unique digital fingerprint of data from any size input. It is one-way, you cannot reverse a hash to get original data. E.g, SHA-256, MD5.

DOMAIN 5: SECURITY OPERATIONS

Security Awareness Training

- Security awareness training helps employees understand cybersecurity risks and learn how to recognize and avoid threats, especially social engineering attacks, which exploit human behavior rather than technical flaws.



What is Social Engineering?

- This is a method attackers use to manipulate people into revealing confidential information or performing actions that compromise security. E g,
- Phishing: Fraudulent emails used to trick users into clicking links or giving up sensitive info. Often looks like messages from trusted sources.
- Tailgating: A person gains physical access to a secure area by following an authorized individual.
- Pretexting: The attacker fabricates a false identity or scenario to steal information.

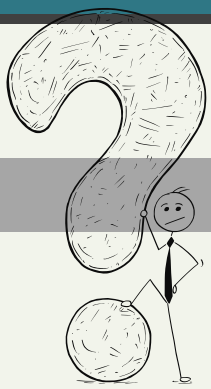
Password Protection

- involves creating, managing, and safeguarding passwords to prevent unauthorized access to systems, accounts, and data.
- Strong, unique passwords combined with MFA significantly reduce the risk of account compromise.

DOMAIN 5: SECURITY OPERATIONS



Quick Check



Question 1: An employee receives a tailored email requesting login credentials. What attack is this?

A. Phishing
B. Spear Phishing
C. Pretexting
D. Tailgating

Answer: B. Spear Phishing, a targeted email attack

Question 2: You see multiple failed login attempts from one IP, and a user reports suspicious activity on their account. What should you do first?

A. Disable affected accounts and block the IP
B. Ignore the alerts and wait.
C. Ask the user to change their password and monitor.
D. Delete the alert logs.

Answer: A. Disable affected accounts and block the IP.

CONCLUSION/RESOURCES

This crash course equips you with ISC2 CC Exam Essentials.

- Click here to know how I passed the CC Exam
- Click here to access my drive containing the full couse and 600 practice quiz

CC Champions Community (C4)

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Ask Me Anything



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